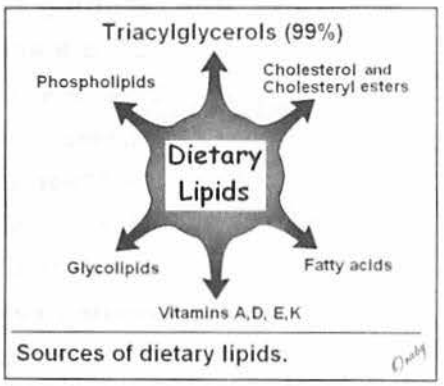
Chapter 3 *Lipids Metabolism*

I. *INTRODUCTION:*

On average, an adult human eats about 100-150 grams lipids per day. The main lipids in diet are triacylglycerols, TG (99%). Diet also contains some phospholipids, cholesterol and fat soluble vitamins.



Sources of dietary lipids.

II. *IMPORTANCE OF LIPIDS IN DIET:*

A. Lipids are one of the main sources of ***energy*** in the body.

B. Lipids supply the body with ***essential fatty acids.***

C. Lipids supply the body with ***fat soluble vitamins.***

D. Lipids make diet ***palatable***.

III. *DIGESTION OF LIPIDS:*

A. **Digestion of triacylglycerols**:

Triacylglycerols are digested by a group of enzymes. These are lingual, gastric, pancreatic and intestinal lipase enzymes.

1. Lingual lipase:

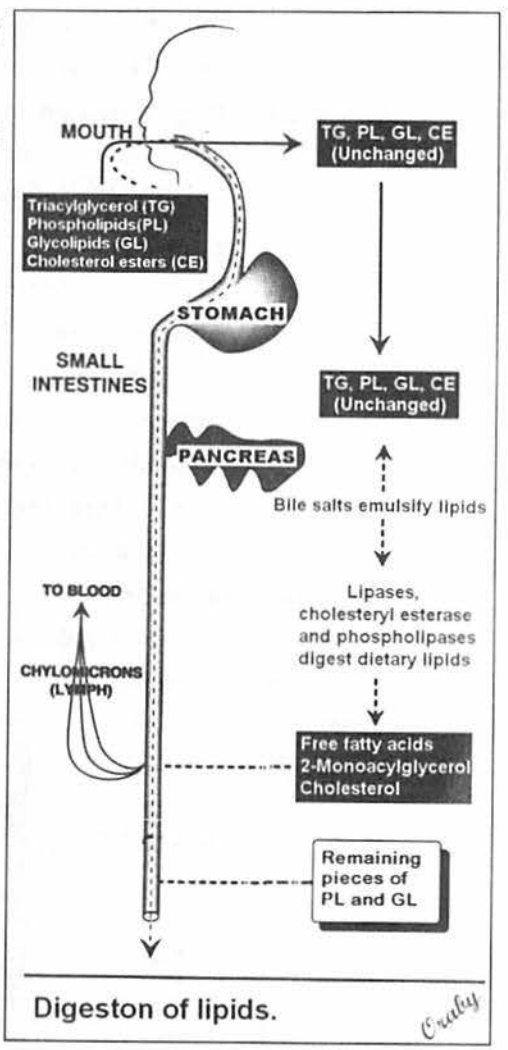
a) Secreted by Ebner's glands on the dorsal surface of the tongue.

b) Because food remains for a short time in the mouth, digestion of triacylglycerols by lingual lipase is minimal.

2. Gastric lipase:

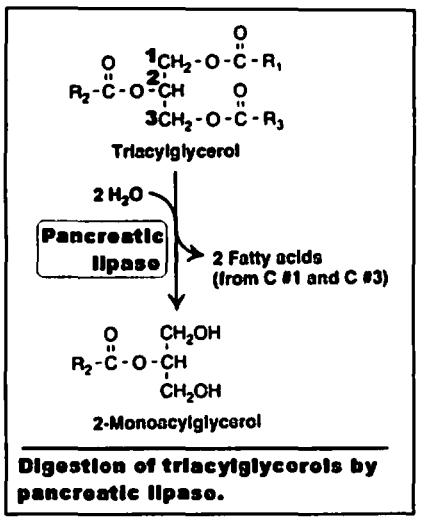
a) Optimum pH for gastric lipase is 7. Thus it cannot act in adult stomach (pH: 1-2).

b) Gastric lipase may be of value in infants stomach (pH: 5). It acts on milk fats.



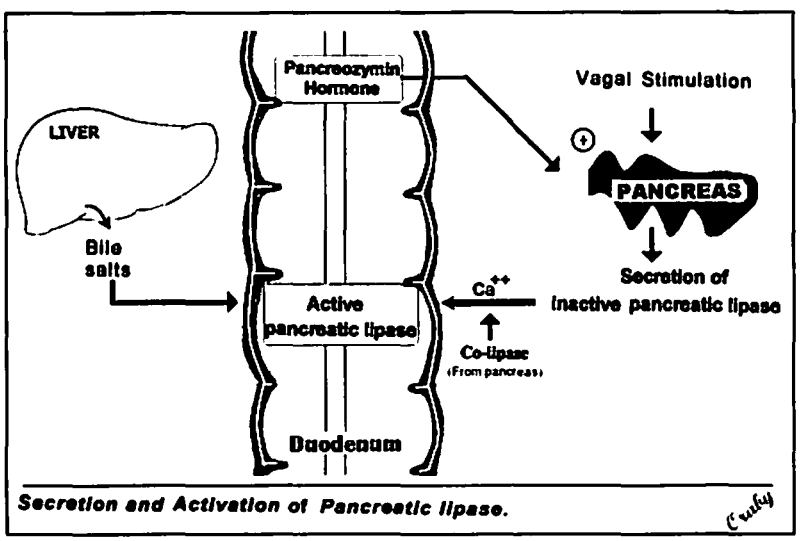
Digeston of lipids.

3. **Pancreatic lipase:**
   - *a)* It is the most important lipase in digestion of triacylglycerols.
   - *b)* It attacks the primary ester bonds of TG (position 1 & 3), hydrolyzing them into fatty acids and 2-monoacylglycerols.
   - *c)* **The resulting 2-monoacylglycerols will undergo:**
     1) 72% are absorbed as such.
     2) 28% are converted into 1-monoacylglycerols by isomerase enzyme which are then :
        - i- Absorbed as 1-monoacylglycerols (6%)
        - ii- Hydrolyzed by pancreatic lipase into glycerol and fatty acids (22%) which are then absorbed.
   - *d)* **Pancreatic lipase is secreted as inactive enzyme:**
     1) Its secretion is stimulated by pancreozymin hormone (secreted by the duodenum) and vagus nerve stimulation.
     2) It is activated in the duodenum by bile salts, calcium ions and colipase (protein produced by the pancreas).
     3) The presence of emulsifying agents as bile salts and phospholipids is important for action of pancreatic lipase. Emulsification means breakdown of large fat globules into small ones. This increases the surface area of lipids exposed to lipase enzyme.



Digestion of triacylglycerols by pancreatic lipase.

4. **Intestinal lipase:**
   - *a)* It acts on 1-monoacylglycerols converting them into glycerol and free fatty acids.

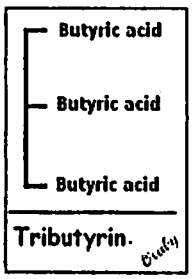


Secretion and Activation of Pancreatic lipase.

The end products of triacylglycerols digestion are:
- 72%: 2-monoacylglycerols + Fatty acids.
- 6% : 1-monoacylglycerols.
- 22%: Glycerol and fatty acids.

B. Digestion of cholesterol and tributyrin:

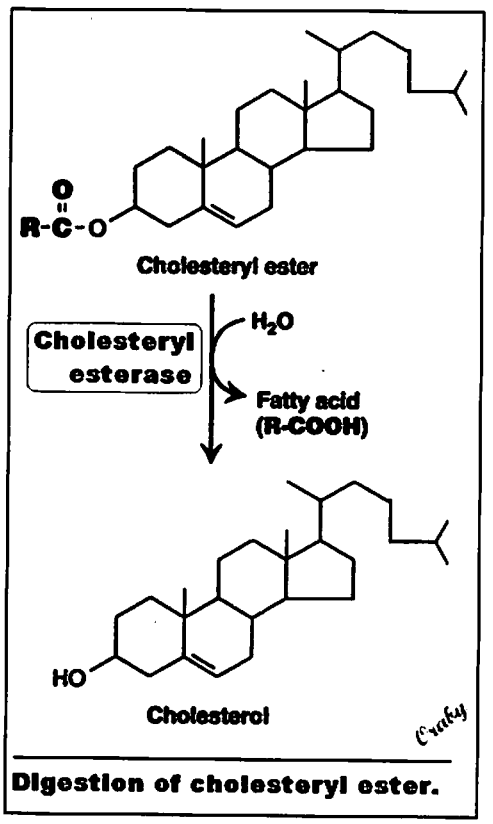
1. Cholesterol itself undergoes no digestion and absorbed as such.
2. Cholesteryl esters are digested by pancreatic cholesterol esterase enzyme into cholesterol and fatty acids.
3. Cholesterol esterase enzyme can also act on tributyrin (triacylglycerols containing 3 butyric acids) hydrolyzing them into glycerol and 3 butyric acid molecules. Milk is rich in tributyrin.



Tributyrin.

C. Digestion of phospholipids:

1. Phospholipids may be absorbed as such or digested by phospholipase enzymes (A_1, A_2 (B), C and D)
2. They act on phospholipids hydrolyzing them into fatty acids, glycerol, phosphate and nitrogenous bases.

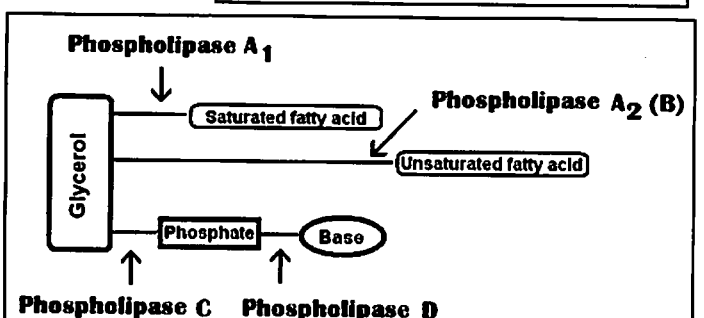


Digestion of cholesteryl ester.

IV. *ABSORPTION OF LIPIDS:*

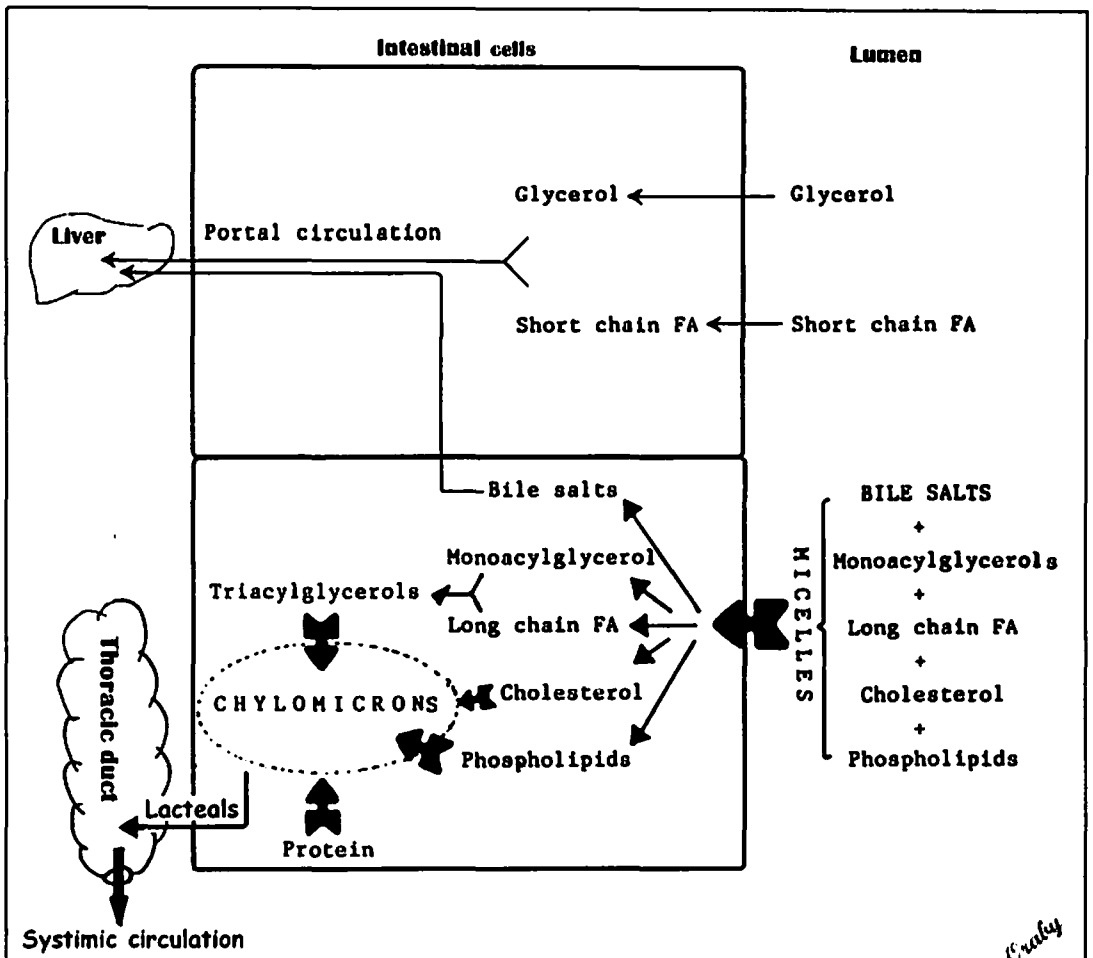
A. Mechanism:

1. **The end products of lipids digestion** are monoacylglycerols, fatty acids [short chains & long chains], glycerol, phospholipids, and cholesterol. They are absorbed from **jejunum and ileum.**
2. Short chain fatty acids (less than 12 carbons) and glycerol are water-soluble and pass via portal circulation to the liver.
3. Other lipids are water insoluble. They combine with bile salts to form a water-soluble complex called **micelles**, which enter the mucosal cells.



Digestion of phospholipids.

4. Bile salts are reabsorbed to the liver again **(enterohepatic-circulation).**
5. Long chain fatty acids are **activated** in the mucosal cells and combine with mono & diacylglycerols to form triacylglycerols again.
6. Triacylglycerols, phospholipids and cholesterol are bound to a protein called apolipoprotein B_{48} to form chylomicrons which enter the circulation through lymphatic vessels.
7. In the blood; chylomicrons are bound with other 2 proteins (apolipoproteins E & C).
8. Chylomicrons will be metabolized as discussed later.



Digestion of lipids.

B. Errors of lipids digestion and absorption:

1. Steatorrhoea :

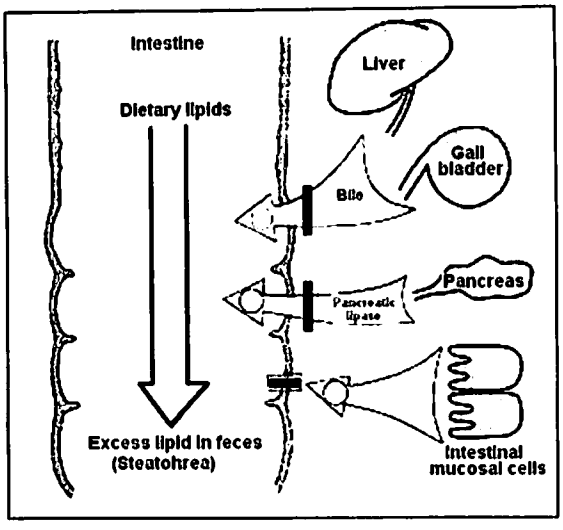
a) **Definition:** It is a condition in which fat content of the stool is abnormally increased. Normally it is less than 5 grams per day.

b) **Causes:** Steatorrhoea results from deficiency of any factor essential for digestion or absorption of lipids as pancreatic lipase, bile salts or healthy intestinal mucosa:

1) **Deficiency of pancreatic lipase :**

i- It is due to **pancreatic duct obstruction, pancreatitis** (inflammation of pancreas) or **Zollinger Ellison disease** (where the high acid content of stomach enters the duodenum and inactivates lipase enzyme.

ii- The fat in stool is undigested. There is no loss of fat-soluble vitamins because vitamins need no digestion.



2) **Deficiency of bile salts :**

i- It occurs due to **bile duct obstruction** (by stone or tumor) or **liver disease** (as in cirrhosis).

ii- The fat in stool is digested and there is loss of fat soluble vitamin (as they need bile salts for absorption).

3) **Defective cells of intestinal mucosa.**

2. Chyluria (milky urine):

a) It is the presence of fat (chylomicrons) in urine after fatty meal.

b) It is due to abnormal connection between the lymphatic drainage of the intestine and the urinary system e.g. as in elephantiasis.

V. *FATE OF ABSORBED LIPIDS:*

A. <u>After fatty meal,</u> plasma shows a milky appearance. This is due to venous blood contains excess chylomicrons after absorption.

1. Excess chylomicrons stimulate mast cells to produce heparin. Heparin then stimulates the lining epithelium of blood vessels of heart, lungs, spleen and adipose tissue to produce an enzyme called: lipoprotein lipase (plasma clearing factor).

2. <u>Lipoprotein lipase enzyme</u> (after being activated by a protein called apolipoprotein C II) will act on triacylglycerols of chylomicrons, converting them into glycerol and free fatty acids.



B. Glycerol and fatty acids are taken up by different tissues for the following fate:

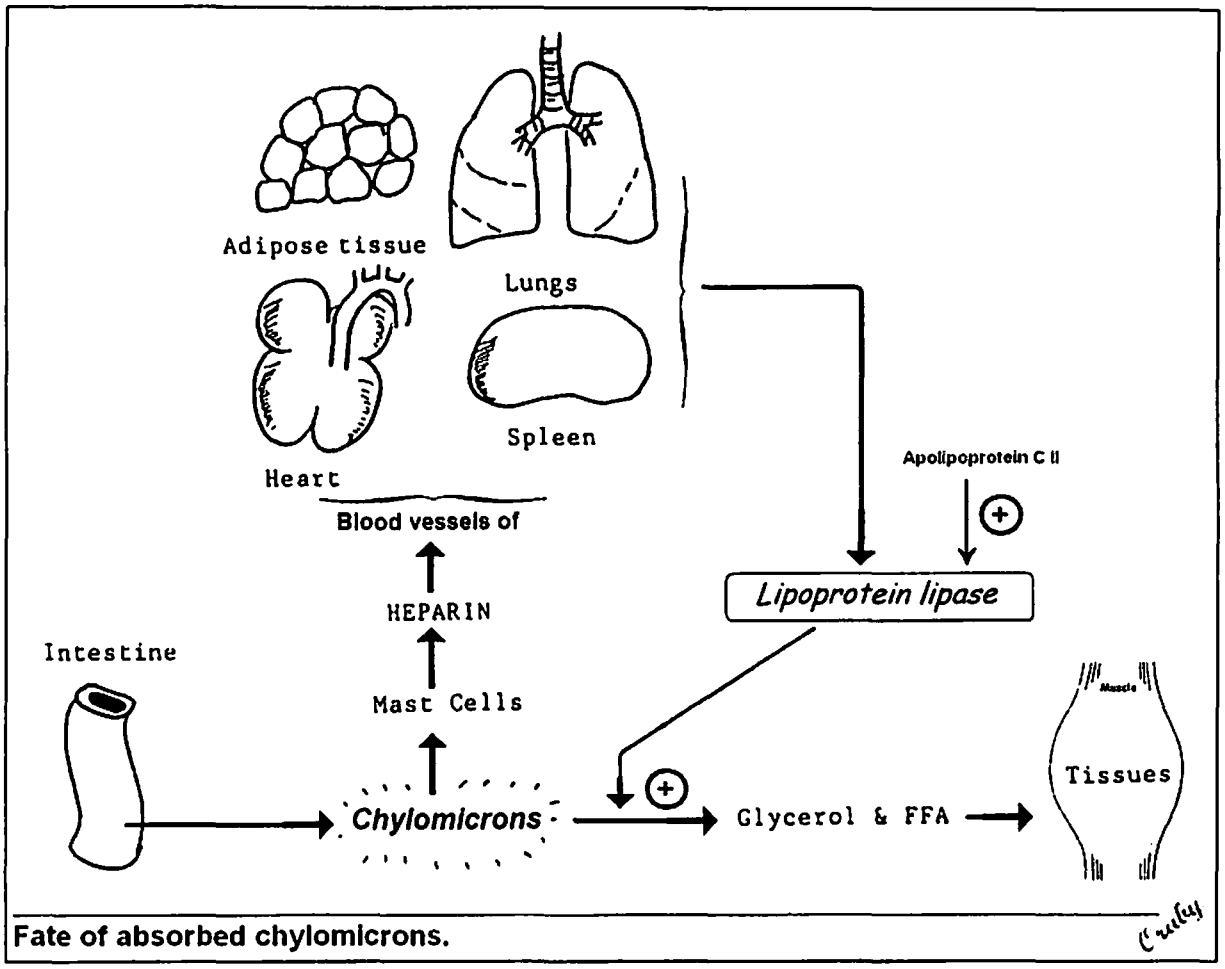
1. **Formation of depot fat (adipose tissue).**
   a) Adipose tissue is formed mainly of triacylglycerols.
   b) It is present in fat cells of adipose tissue.

2. **Oxidation for production of energy:**

3. **Glucose formation by gluconeogenesis:**
   a) Glycerol → Glucose.

4. **Synthesis of biologically active compounds:** e.g. different steroids and eicosanoids (prostaglandins, prostacyclin, thromboxanes and leuckotrienes).

5. **Synthesis of tissue fats (structural cellular fats):** They include mainly phospholipids and cholesteryl esters.



Fate of absorbed chylomicrons.

STORAGE AND MOBILIZATION OF LIPIDS

I. ***Types of body lipids:*** Body lipids are of 4 types; tissue lipids, adipose tissue (depot fat), plasma lipoproteins and bone marrow lipids.

A. **Tissue lipids:**

1. These lipids enter in the structure of body cells as cell membrane and mitochondria.
2. Tissue lipids are never oxidized to give energy.

B. **Adipose tissue (depot fat):** It is of 2 types white and brown.

1. White adipose tissue:

a) **Composition:**

1) Triacylglycerols: main content that contain saturated and unsaturated fatty acids. This makes depot fat in a fluid state at body temperature.
2) Little phospholipids and cholesterol.

b) **Site:**

1) Under skin (subcutaneous fat) and in the breast.
2) Around important organs e.g. kidney.
3) In the omentum and mesentry.

c) **Sources:**

1) Absorbed fat.
2) Carbohydrate (by lipogenesis).

d) **Functions:** depot fat is important for:

1) **Energy production:** During fasting, the triacylglycerols stored in depot fat provide the body with free fatty acids that are oxidized to give energy.
2) **Fixation of some organs e.g. kidney.**
3) **Heat insulator around the body.**
4) **Production of vitamin D_3:** Exposure of skin to ultraviolet rays of sun transforms 7-dehydrocholesterol into vitamin D_3.

2. Brown adipose tissue:

a) Certain areas of adipose tissue appear brown in color as they contain high content of mitochondria, cytochromes and well developed blood supply.

b) The brown adipose tissue may have a special function in the production of heat. They contain a protein called **thermogenin.** This protein acts as uncoupler of oxidative phosphorylation ➔ ↓ ATP production and ↑ heat generation.

c) Brown adipose tissue is common in animals exposed to cold atmosphere for warmness. It is little in human, especially in obese persons.